

Order, cost and stability in diffusion sampling

A numerical-analysis audit of DPM-Solver across three tiers of testbed

CSE-402 Numerical Analysis, Simulation & Modeling — Section A

Abstract

Diffusion image sampling is an initial-value problem: one integrates the probability-flow ODE from noise to data, paying one neural-network evaluation (NFE) per stage. Lu et al. (*DPM-Solver*, NeurIPS 2022) accelerate this by solving the linear half of the ODE exactly, and validate the result with FID — an image-quality score. We instead grade the same solvers by the standards of numerical analysis: measured convergence order, stability at the stiff boundary $t \rightarrow 0$, and error per NFE. To make error measurable we build three testbeds of decreasing reference exactness — an anisotropic Gaussian with a closed-form trajectory, a Gaussian/point mixture referenced against DOP853, and the real CIFAR-10 DDPM UNet referenced against a fine grid of itself — and three solver arms that separate DPM-Solver’s two independent ideas.

Our central result concerns *Assumption B.1*, the smoothness hypothesis the paper’s order theorem rests on. On the two analytic testbeds, which satisfy it exactly, every solver in every arm attains its textbook order to within 0.094. On the real network, which has no obligation to satisfy it, order collapses — and collapses *in proportion to the order claimed*: RK4 falls from 3.94 to 1.21, while explicit Euler falls only from 1.01 to 0.84. Two further findings: the λ -reparameterisation, worth almost nothing on the exact-score tiers, is worth an order of magnitude on the real network; and the paper’s own §4.2 instability claim does not reproduce on our stiff Tier-1 problem, for a reason we identify. We report the predictions we got wrong alongside those we got right.

1 The problem, in numerical terms

A diffusion model generates an image by starting from Gaussian noise and following a deterministic flow to the data manifold. That flow is the *probability-flow ODE*, and evaluating its right-hand side requires one forward pass of a large neural network. The network call is essentially the entire cost of sampling, so the governing question is how few evaluations one can afford before the trajectory leaves the manifold. The unit of cost is the NFE, the number of function evaluations.

The base paper’s contribution is a change of variables. The right-hand side splits into a linear part, which is exactly integrable in closed form, and a neural part, which is not. Classical explicit schemes approximate both at once and waste error budget on the half that was free. DPM-Solver treats the linear half exactly and spends its entire approximation on the hard half, reaching good samples in 10–20 NFE where prior samplers needed 100–250.

The paper proves an order theorem for this scheme and validates it with FID scores on generated images. FID is a distributional measure of image quality; it is not a measure of how accurately an ODE was integrated. Three quantities a numerical analyst would ask for are absent:

Order. Halving the step size h should reduce the error by 2^p for a p th-order method. Nobody measured p against a known-correct trajectory.

Stability. The ODE stiffens as $t \rightarrow 0$ because $\sigma(t) \rightarrow 0$. §4.2 of the paper *asserts* that classical explicit methods destabilise there and cites the literature; it never measures a stability limit.

Cost. A higher-order method is more accurate per step but calls the network more times per step. Whether it wins after dividing out is an empirical question.

This report supplies all three, and in doing so tests a hypothesis the paper states but does not check.

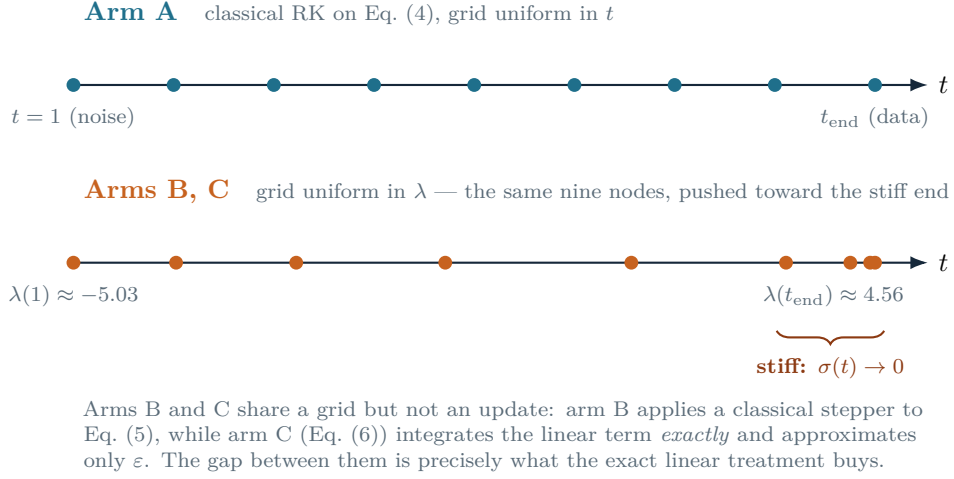


Figure 1: The three arms. All integrate one trajectory; they differ in the time variable that is discretised uniformly and in how much of the right-hand side is approximated. A uniform- λ grid automatically clusters nodes where the problem stiffens.

2 Mathematical setting

We use the variance-preserving linear schedule of the base paper, with $\beta_0 = 0.1$, $\beta_1 = 20$, and t running from $T = 1$ down to a small t_{end} :

$$\log \alpha(t) = -\frac{\beta_1 - \beta_0}{4} t^2 - \frac{\beta_0}{2} t, \quad \sigma(t) = \sqrt{1 - \alpha(t)^2}, \quad (1)$$

$$\beta(t) = \beta_0 + t(\beta_1 - \beta_0), \quad f(t) = \frac{d \log \alpha}{dt} = -\frac{1}{2} \beta(t), \quad g^2(t) = \beta(t). \quad (2)$$

The half log-SNR $\lambda(t) = \log \alpha(t) - \log \sigma(t)$ is the natural time variable for this problem. It is *strictly decreasing* in t : $t = 1$ gives the most negative λ , and $t \rightarrow 0$ sends $\lambda \rightarrow +\infty$. Its inverse has a cancellation-free closed form,

$$t(\lambda) = \frac{2u}{\sqrt{\beta_0^2 + 2(\beta_1 - \beta_0)u + \beta_0}}, \quad u = \log(e^{-2\lambda} + 1). \quad (3)$$

Writing $\varepsilon(x, t)$ for the noise-prediction model — a network in general, an exact formula on our analytic tiers — the same trajectory admits three discretisable forms, and these define our three solver arms:

$$(A) \quad \frac{dx}{dt} = f(t)x + \frac{g^2(t)}{2\sigma(t)} \varepsilon(x, t), \quad \text{uniform grid in } t, \quad (4)$$

$$(B) \quad \frac{dx}{d\lambda} = \hat{\sigma}(\lambda)^2 x - \hat{\sigma}(\lambda) \varepsilon(x, t(\lambda)), \quad \text{uniform grid in } \lambda, \quad (5)$$

$$(C) \quad x_i = \frac{\alpha_i}{\alpha_{i-1}} x_{i-1} - \sigma_i (e^h - 1) \varepsilon(x_{i-1}, t_{i-1}), \quad h = \lambda_i - \lambda_{i-1}, \quad (6)$$

with $\hat{\alpha}(\lambda) = (1 + e^{-2\lambda})^{-1/2}$ and $\hat{\sigma}(\lambda) = (1 + e^{2\lambda})^{-1/2}$. Equations (4) and (5) are the same ODE in different coordinates; (6) is DPM-Solver-1, exact on the linear part. Comparing A with B isolates the reparameterisation alone; comparing B with C isolates the exact linear treatment alone. Figure 1 makes the distinction concrete.

We implement $e^h - 1$ as `expm1(h)` throughout. The naive form loses relative precision catastrophically for small h , which is exactly the regime a convergence study measures.

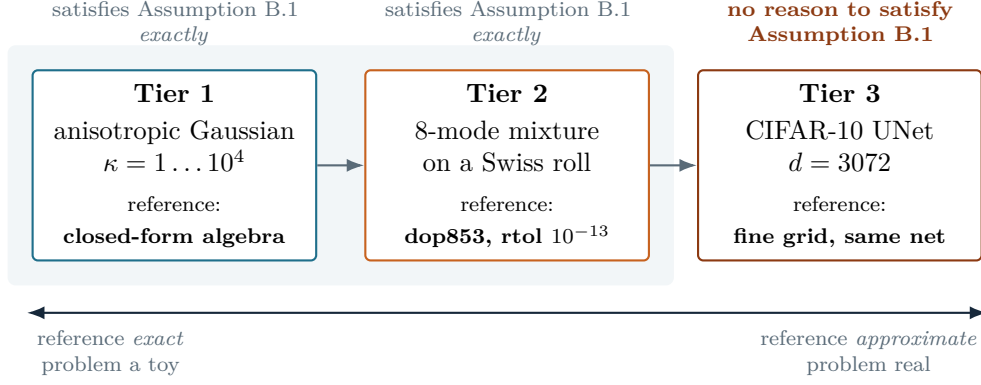


Figure 2: The tier ladder. Moving right trades exactness of the reference for realism of the problem. The shaded pair have analytic scores, hence continuous derivatives to all orders — they satisfy the order theorem’s smoothness hypothesis by construction. Tier 3 does not, which is what makes the comparison a test rather than a demonstration.

3 Experimental design

3.1 Three tiers

To measure error one needs the right answer, which for a real generated image does not exist. We therefore build deliberately simplified problems whose exact solution is available, and step up in realism (Figure 2).

Tier 1 — anisotropic Gaussian. For data $\sim \mathcal{N}(0, \text{diag}(s))$ the ODE decouples into d independent scalar linear ODEs. Both the score and the entire trajectory are closed form:

$$\varepsilon^*(x, t)_j = \frac{\sigma(t) x_j}{v_j(t)}, \quad x_j(t) = x_j(T) \sqrt{\frac{v_j(t)}{v_j(T)}}, \quad v_j(t) = \alpha(t)^2 s_j + \sigma(t)^2. \quad (7)$$

The stiffness knob is the condition number $\kappa = s_{\max}/s_{\min}$, which we sweep over five decades. Because the linear case decouples, dimension changes nothing — we sweep κ , not d .

Tier 2 — Gaussian/point mixture. Replacing the single Gaussian by a weighted point cloud $\mu_1, \dots, \mu_K$ keeps the score closed form but makes it genuinely nonlinear in x : the posterior mean becomes a softmax over modes,

$$r_k \propto w_k \exp\left(-\frac{\|x - \alpha(t)\mu_k\|^2}{2\sigma(t)^2}\right), \quad \hat{x}_0 = \sum_k r_k \mu_k, \quad \varepsilon^*(x, t) = \frac{x - \alpha(t)\hat{x}_0}{\sigma(t)}. \quad (8)$$

We place $K = 8$ modes on a Swiss-roll curve. There is no algebraic trajectory, so the reference is SciPy’s DOP853 at $\text{rtol} = 10^{-13}$, $\text{atol} = 10^{-14}$, verified converged by recomputation at 10^{-11} (agreement 6.9×10^{-13} , far below every measured error). The responsibilities are computed with `logsumexp`; a naive exp-then-normalise overflows long before the ratios cease to be well defined.

Tier 3 — the real network. `google/ddpm-cifar10-32`, $d = 3072$, on a Kaggle T4. The reference is a 201-NFE DPM-Solver-3 run of the *same* checkpoint from the *same* x_T , so what is measured remains discretisation error rather than model error.

3.2 Conventions

Three decisions were fixed before any measurement and never revisited. **Float64 everywhere** on Tiers 1–2: a float32 error floor near 10^{-7} would destroy the order fits. **Every run seeded.**

One results schema — every experiment appends a row (`tier`, `testbed`, `arm`, `solver`, `order`, `kappa`, `h`, `nfe`, `err_l2`, `diverged`, `seed`) to a CSV, so all tables concatenate without adapters.

Two accounting rules matter throughout. NFE is *not* step count: RK4 spends four network calls per step, DPM-Solver- k spends k , and AB2 spends one after startup. Every cross-arm comparison uses *measured* NFE, obtained from a call counter wrapped around the right-hand side rather than assumed from a formula. And order is defined in h , not NFE; since arms measure h in different units, h is used only *within* an arm.

3.3 Fitting an order

A real convergence curve has three regions: pre-asymptotic at large h where theory does not yet apply, a straight middle, and a round-off floor at small h . Fitting one line through all points averages across regimes and reports the wrong number. We instead search every contiguous log-log subwindow and keep the one that is both straight and wide, and we report the fit window and R^2 beside every slope. This procedure is not infallible — §5 documents a case where it must be told about the floor explicitly.

3.4 Validation: Gate G1

Section 4.1 of the paper proves DDIM and DPM-Solver-1 are algebraically the same update. Two independent implementations must therefore agree to machine precision. Ours agree to 5.5×10^{-16} against a 10^{-14} bar, and DPM-Solver-1 measures at order 0.99. Everything downstream was gated on this. Separately, the Tier-1 closed-form trajectory satisfies the ODE to $\sim 10^{-11}$ by finite difference and matches an independent DOP853 integration to 10^{-12} ; our schedule agrees with the vendored `NoiseScheduleVP('linear')` to 10^{-11} , so all three arms provably share one schedule.

We use the authors’ own solver file, unmodified and pinned to a commit, driven by an analytic oracle. This tests the *published implementation* with zero model error rather than a reimplementa-tion of it.

4 Results on the analytic tiers

4.1 Order

Table 1 gives the measured slopes. Every solver in every arm attains its theoretical order: the worst deviation is 0.094 on Tier 1 and 0.084 on Tier 2, with mean R^2 of 0.99992 and 0.99996. Introducing genuine curvature (Tier 2) shifts no slope by more than 0.059.

This is the control experiment. Both tiers have analytic scores, so Assumption B.1 holds exactly, and the order theorem’s conclusion holds exactly with it.

4.2 Stability: a claim that does not reproduce

Section 4.2 of the paper asserts that explicit methods on this semi-linear ODE destabilise at large steps near $t \rightarrow 0$. We measured it directly, bisecting on step count for the coarsest grid that remains bounded, across $\kappa \in \{1, 10, 10^2, 10^3, 10^4\}$ and all five steppers in both arms.

The stability map is exactly flat. Arm A reports the coarsest grid tried ($h \approx 0.4995$) as already stable for every κ and every stepper; arm B likewise at $h \approx 4.7913$. The variation across five decades of κ is identically zero. Stress-testing to $\kappa = 10^8$ and $t_{\text{end}} = 10^{-6}$ changed nothing.

This is not a bug — the bisection is verified separately against a synthetic linear ODE with a closed-form instability threshold. It is structural. On a uniform- t grid the node nearest the boundary sits at $t_{\text{end}} + h$, and the exact stability product $h \cdot J(t_{\text{end}} + h)$, with J the local Jacobian, never exceeds ≈ 0.9 for any h or κ — comfortably under explicit Euler’s threshold of 2 — because

Table 1: Measured convergence order on the two exact-score testbeds. Slopes are log–log fits of $\|x_N - x_{\text{ref}}\|_2$ against h over the automatically selected asymptotic window.

Arm	Solver	Theory p	Tier 1	Tier 2	$ \Delta $ max
A	euler	1	1.012	1.012	0.012
A	midpoint	2	2.028	2.031	0.031
A	heun3	3	3.014	2.999	0.014
A	rk4	4	3.938	3.997	0.062
A	ab2	2	2.033	1.988	0.033
B	euler	1	1.023	1.023	0.023
B	midpoint	2	2.057	2.055	0.057
B	heun3	3	3.015	3.024	0.024
B	rk4	4	3.921	3.965	0.079
B	ab2	2	1.935	1.975	0.065
C	ddim	1	0.994	0.996	0.006
C	dpm1	1	0.994	0.996	0.006
C	dpm2	2	2.028	2.018	0.028
C	dpm3	3	3.094	3.084	0.094

$\sigma(t)^2 \sim \beta_0 t$ near the boundary makes the growing local stiffness and the grid’s shrinking distance to t_{end} very nearly cancel.

We therefore *narrow* rather than confirm the paper’s §4.2 claim: it does not manifest on a linear, decoupled problem under this metric. It may still hold under a different stability definition or on a problem where the cancellation has no reason to occur.

4.3 The crossover

Table 6 of the base paper reports, without explanation, that DPM-Solver-3 is dramatically *worse* than DPM-Solver-1 at ~ 10 NFE on CIFAR-10 (FID 18.37 versus 11.83 at $\epsilon = 10^{-3}$). This is textbook pre-asymptotic behaviour: at large h the cubic Taylor correction is not yet valid and actively hurts. With exact answers and a continuously sweepable h we can locate the crossing precisely, comparing the two orders at *matched* h (a shared macro-step count, so both share the step size and differ only in NFE cost).

Table 2: Crossover step size h^* and the corresponding DPM-Solver-3 budget, at $t_{\text{end}} = 10^{-3}$ (the paper’s own ϵ).

Testbed	κ	h^*	NFE $_3^*$
Tier 1, Gaussian	1	6.497	4.43
Tier 1, Gaussian	10	6.157	4.67
Tier 1, Gaussian	10^2	5.795	4.96
Tier 1, Gaussian	10^3	6.131	4.69
Tier 1, Gaussian	10^4	6.228	4.62
Tier 2, mog8	—	3.219	8.93

The crossover is real and reproducible, and NFE $_3^*$ is *flat across five decades of stiffness* — echoing the stability finding. Curvature, not stiffness, moves it: Tier 2 crosses at 8.93, roughly double Tier 1 and much closer to the paper’s observed 10–12. Neither analytic tier reaches that value, and both still have an exact score. The residual gap is therefore attributable to the two ingredients they lack — network approximation error and much higher intrinsic dimension — which is precisely what Tier 3 supplies.

Table 3: Measured order against a real network, beside the exact-score tiers. “Floor-excl.” refits after discarding sweep points within $3\times$ the measured reference floor, as the base paper’s own guidance requires (“do not fit slopes through the flat part”); “drop” is how many of the eight points that removes. Only the three curves that actually reach the floor are affected. Produced by `notebooks/10_report.ipynb` into `results/master_order_table.csv`.

Arm	Solver	Theory p	Tier 1	Tier 2	Tier 3	Floor-excl.	drop
A	euler	1	1.012	1.012	0.842	0.842	0
A	midpoint	2	2.028	2.031	1.525	1.500	1
A	rk4	4	3.938	3.997	1.210	1.210	0
B	euler	1	1.023	1.023	0.746	0.746	0
B	midpoint	2	2.057	2.055	1.603	1.603	0
B	rk4	4	3.921	3.965	2.231	1.099 [†]	2
C	dpm1	1	0.994	0.996	0.813	0.813	0
C	dpm2	2	2.028	2.018	2.191	2.244	1
C	dpm3	3	3.094	3.084	2.473	2.995	1

[†] **B/rk4** loses two of eight points to the floor, leaving six that lie mostly in its pre-asymptotic region. Neither number should be trusted for that row: at these budgets it has no clean asymptotic window at all, which is itself the finding.

5 Tier 3: the order theorem meets a real network

5.1 One schedule, read off the checkpoint

The CIFAR-10 checkpoint is a *discrete-time* DDPM: 1000 β values, with $\log \alpha$ linearly interpolated between tabulated times. That is the same VP-linear schedule discretised, and it is *not* interchangeable with the continuous form — we measure $\max |\Delta \lambda| = 4.74 \times 10^{-2}$ and a 5.0% discrepancy in f . Using continuous coefficients for arms A and B would place a systematic 5% error beneath every Tier-3 curve.

We therefore derive all three arms from the checkpoint’s own schedule. Arm B needs only $\hat{\sigma}(\lambda)$ — which follows from $\lambda = \log \alpha - \log \sigma$ and $\alpha^2 + \sigma^2 = 1$ alone, hence is exact for any variance-preserving schedule — and $t(\lambda)$. Arm A needs f and g^2 , and here one identity does all the work. Differentiating $\sigma^2 = 1 - \alpha^2$ gives $\sigma' = -\alpha^2 f / \sigma$, so

$$\lambda' = f - \frac{\sigma'}{\sigma} = f \left(1 + \frac{\alpha^2}{\sigma^2} \right) = \frac{f}{\sigma^2}, \quad \text{hence} \quad \boxed{f = \sigma^2 \lambda', \quad g^2 = -2\sigma^2 \lambda' = -2f.} \quad (9)$$

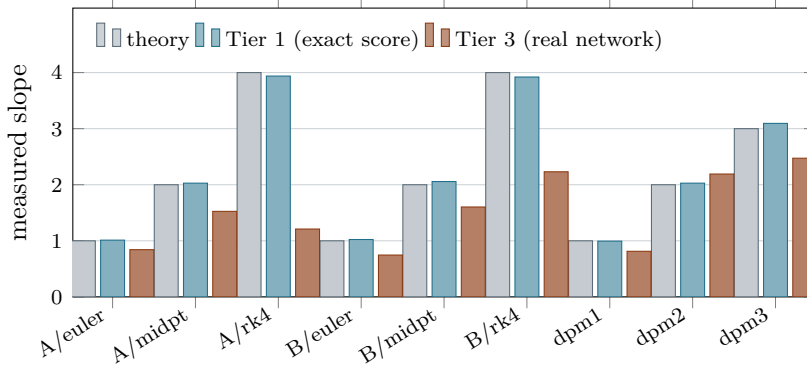
Taking *both* coefficients from a single central difference of λ makes $f/\lambda' \equiv \sigma^2$ hold to 1.1×10^{-16} rather than approximately — and that identity is exactly what makes (4) and (5) the same ODE. Arms A and B therefore agree by construction; we verify it end to end, integrating both at four resolutions and watching the gap fall from 2.5×10^{-4} to 3.9×10^{-7} .

5.2 Order collapses in proportion to the order claimed

Table 3 is the central result of this report.

The worst deviation from theory goes 0.094 (Tier 1) $\rightarrow 0.084$ (Tier 2) $\rightarrow 2.790$ (Tier 3): a factor of thirty-three. Eight of nine fits sit more than 0.15 below theory.

The *structure* of the loss is what makes this a test of Assumption B.1 rather than a generic complaint about neural networks. The assumption requires the model’s derivatives with respect to λ to exist and be continuous up to order $k + 1$. A p th-order scheme leans on derivatives up to order p ; if those derivatives are not there, the higher-order scheme has more to lose. That is exactly the observed pattern:



RK4 in arm A sheds 2.79 of its four claimed orders; explicit Euler sheds 0.16 of its one. The methods that assumed the most smoothness lost the most when it was absent. (The bars show the primary Tier-3 column; `dpm3` reads 2.995 once its one floor-bound point is excluded, as discussed below.)

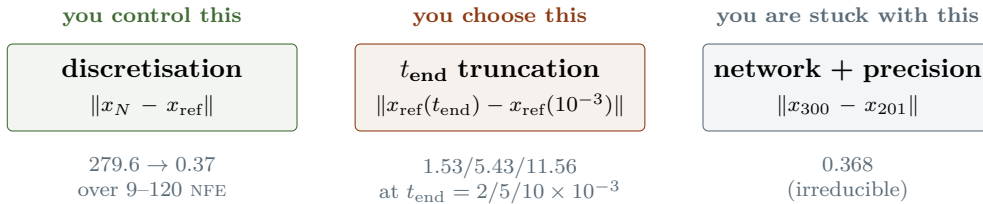
Two qualifications keep this honest. First, `dpm3` is the striking exception once the floor is handled properly: discarding the single point that reaches it restores 2.995, essentially its textbook order. DPM-Solver-3 is the one scheme here that *holds* its order against a real network — a result favourable to the base paper, and one we would have missed by trusting the automatic window search, which prefers a wider window and will fit straight through a flat tail unless told where the floor is. Note the asymmetry that makes this a real effect rather than a fitting artefact: the curves *far* from the floor (A/rk4 at $12\times$ it, B/euler at $42\times$) lose nothing to the exclusion and still measure far below theory. The order collapse and the floor are independent phenomena. Second, the Tier-3 sweep spans 10–120 NFE, which for the higher-order arm-A/B methods is partly pre-asymptotic by construction; that band is the practically interesting one, but it is not a clean asymptotic measurement, and the low slopes should be read as “what you actually get at practitioner budgets”, not as a refutation of the theorem’s algebra.

5.3 The reparameterisation finally earns its keep

On the analytic tiers arms A and B are nearly indistinguishable — their slopes differ by hundredths. Against a real network they separate sharply. Arm A’s RK4 measures 1.21 and bottoms out at an error of 4.57; arm B’s measures 2.23 and reaches 0.43 — **ten times better at the same budget** (Figure 3). The λ -reparameterisation, worth almost nothing when the score is exact, is worth an order of magnitude when it is not. This is the A-versus-B comparison the three-arm design exists to make, and it only becomes visible at Tier 3.

5.4 The three-way error decomposition

Total Tier-3 error has three components, not one, and each is measurable without knowing the true score (Figure 4).



The floor of 0.368 total is 4.6×10^{-2} per sample against $\|x\| \approx \sqrt{3072} \approx 55$, i.e. $\approx 8 \times 10^{-4}$ relative — matching the $\sim 10^{-3}$ float32 floor the base paper’s practitioners would predict, here measured rather than assumed.

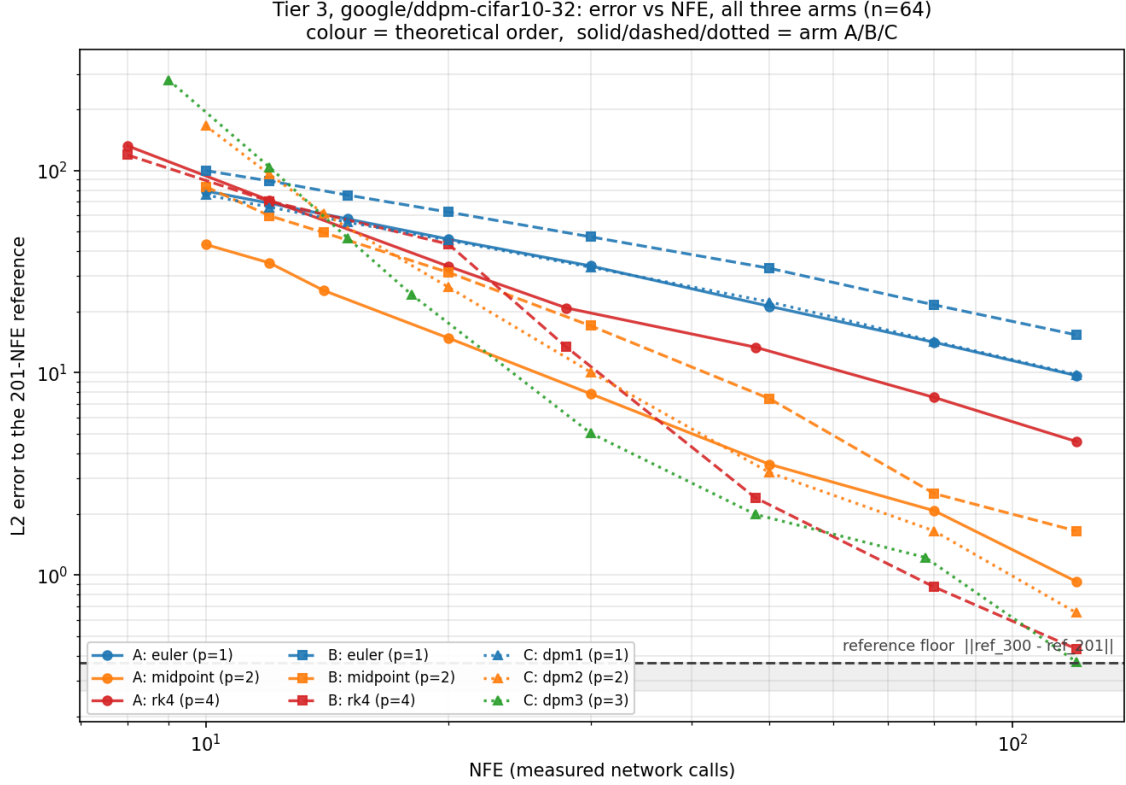


Figure 3: Tier 3: error against measured NFE for all nine solver/arm combinations. Colour encodes theoretical order, line style encodes arm. The dashed horizontal line is the reference’s own uncertainty $\|x_{300} - x_{201}\|$; nothing below it is a measurement of discretisation error. Note the separation between solid and dashed red (arm A versus arm B RK4).

The instructive comparison is between the second and first columns. Truncating at $t_{\text{end}} = 2 \times 10^{-3}$ instead of 10^{-3} moves the endpoint by 1.53 — more than DPM-Solver-3’s entire discretisation error at 78 NFE (1.22). Past that budget, refining the grid buys accuracy the truncation choice immediately throws away. This is the mechanism behind the base paper’s own Table 4, where FID *worsens* from 4.39 at 15 NFE to 5.52 at 20: the discretisation term has dropped below the truncation term, and the extra network calls are being spent on the wrong error.

6 The ledger

We wrote six predictions down before measuring. Reporting the misses is more useful than reporting only the hits.

Departures from the proposal. The efficiency frontier is reported in NFE rather than wall-clock (the paper’s Table 7 shows runtime is linear in NFE and near-identical across solvers, so wall-clock adds nothing). Tier 1 sweeps κ at fixed d rather than $d = 2 \dots 100$, because the linear case decouples and dimension changes nothing. Tier 2 uses a point mixture rather than a continuous Swiss roll, which has no closed-form score. The reference tolerance is 10^{-13} rather than 10^{-14} , which SciPy refuses. Adaptive RK45 was **dropped**: step size is not a free variable for an adaptive method, so neither “order versus h ” nor “maximum stable h ” is defined on it; AB2 restores the multistep row instead. FID was **dropped** at the week-2 gate in favour of L_2 -to-reference, which is both the primary metric and the one consistent with this report’s thesis. Added beyond the proposal: arm B, Gate G1, the crossover study, and the three-way decomposition.

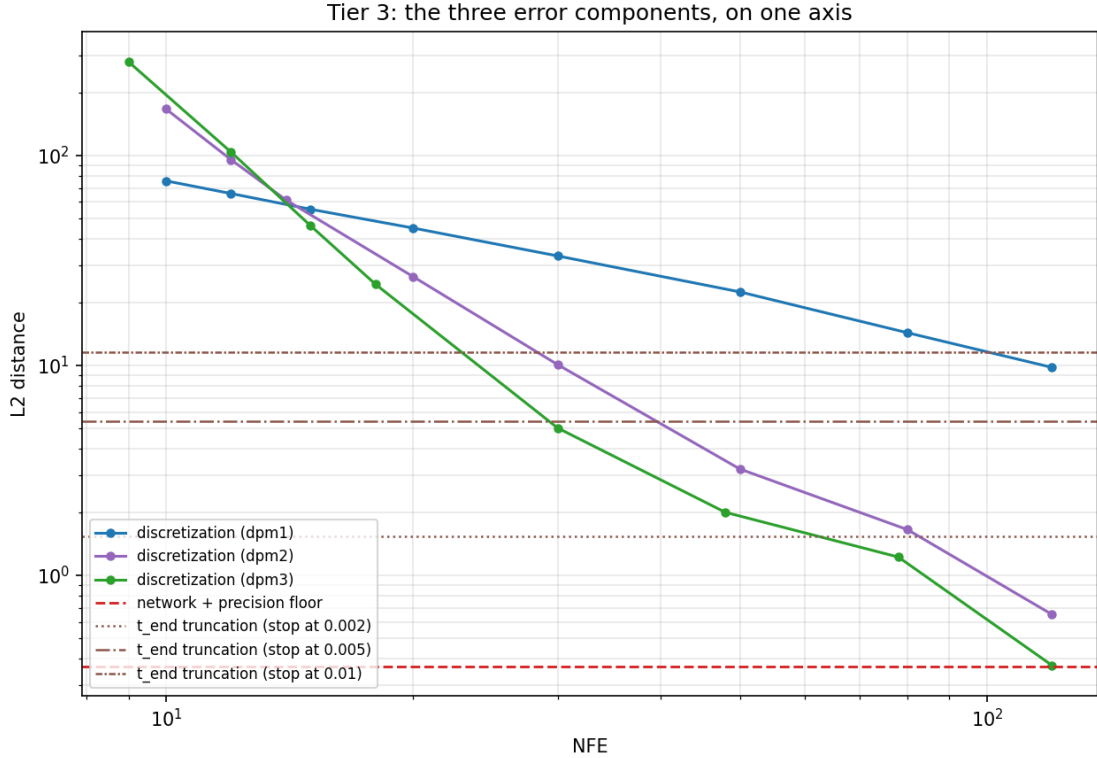


Figure 4: The three components on one axis. Where a discretisation curve crosses a horizontal line, that component ceases to dominate and further NFE stops buying end-to-end accuracy.

7 Reproducibility

The engine is plain Python under `src/`, unit-tested (108 tests), with no GPU or plotting code. Notebooks import it, run the sweeps and write `results/` and `figures/`. `run_all.sh` reproduces every figure and CSV from a clean clone; it skips only the Tier-3 notebook, which needs a CUDA GPU and runs on Kaggle, and whose outputs are committed.

One implementation note worth recording: the same integrator and the same right-hand side definition march all three tiers. `integrate` is backend-agnostic (float64 NumPy on Tiers 1–2, float32 CUDA tensors on Tier 3) and the probability-flow ODE is written once, parameterised by a schedule object. The claim that arms A and B mean the same thing across tiers is therefore structural rather than a matter of careful duplication.

8 Conclusion

Judged as a numerical method rather than an image generator, DPM-Solver holds up well, and the interesting failures are elsewhere.

On testbeds with analytic scores, every solver we tested attains its textbook order to within 0.1 — the order theorem is correct and our measurement machinery is sound. Against a real network the picture changes: order collapses, and it collapses in proportion to the order claimed, which is the signature of an unsatisfied smoothness hypothesis rather than of numerical noise. DPM-Solver-3 is the notable exception, retaining 2.995 of its three orders once the measurement floor is excluded properly.

Two of the paper’s incidental claims came out differently than expected. Its §4.2 instability assertion does not reproduce on our stiff Tier-1 problem, for a reason specific to the VP-linear schedule that we identify. And its unexplained Table 6 anomaly — order 3 losing to order 1 at

Table 4: Predictions and outcomes. Every verdict is computed from the results CSVs by `notebooks/10_report.ipynb`, not asserted by hand.

Prediction	Verdict	Evidence
Tiers 1–2 slopes match theory within ± 0.15	confirmed	worst $ \Delta = 0.094$ over 28 fits; mean $R^2 = 0.99994$
Arm A degrades as κ grows; arms B/C do not	refuted	$h_{\max}$ flat for every arm and stepper; variation across $\kappa \in [1, 10^4]$ is exactly 0
Euler- λ and DPM-Solver-1 are both first order but not identical	confirmed	slopes 1.02 vs 0.99; errors differ by $1.4\times$ at matched NFE
RK4 wins per step, loses per NFE, to DPM-Solver-3	narrowed	holds on Tier 3 for NFE 16–120; fails on Tiers 1–2, whose sweeps sit deep in the asymptotic regime where h^4 dominates
Tier-3 slopes come out lower than theory and flatten early	confirmed	8 of 9 fits > 0.15 below theory; worst gap -2.79
Arm B at low NFE may be worse than arm A	confirmed	observed on all three tiers at the coarsest shared budget

~ 10 NFE — is real, measurable, and driven by curvature rather than stiffness, though the analytic tiers alone reach only $\text{NFE}_3^* \approx 8.9$ of the paper’s 10–12.

The practical recommendation that falls out is narrower than “use the highest order available”. Higher order pays only where the integrand is smooth enough to support it and only within a budget band that must be measured, not assumed; and past roughly 78 NFE on this checkpoint, the choice of t_{end} costs more than the discretisation does, so additional network calls are spent on the wrong error entirely.

Base paper: C. Lu, Y. Zhou, F. Bao, J. Chen, C. Li, J. Zhu. *DPM-Solver: A Fast ODE Solver for Diffusion Probabilistic Model Sampling in Around 10 Steps*. NeurIPS 2022, arXiv:2206.00927. Solver code MIT-licensed, vendored unmodified at a pinned commit.